Sayantan Sikdar

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Summary

M.Tech (IT) candidate and IEEE-published AI/ML engineer who ships end-to-end systems — LLM fine-tuning, durable AI agents, a Fabric lakehouse reconciled to the cent and a dialer verified by 529 tests — and owns the full loop from scoping with stakeholders to deploying and explaining the numbers.

Education

Masters of Technology, *IT*, *Netaji Subhas University of Technology, CGPA: 7.11* **2025 – 2027**

Bachelors of Technology, *CSE (AI / ML)*, *Haldia Institute of Technology, CGPA: 8.65* **2020 – 2024**

Technical Skills

Programming Languages: Python, TypeScript, SQL, Java

LLMs & GenAI: LoRA/QLoRA Fine-Tuning, HuggingFace Transformers, Hybrid RAG (BM25 + Dense, RRF), Embeddings, LanceDB, Agentic Workflows, LLM Evaluation & Benchmarking, Pydantic AI, Claude API, Ollama, Prompt Engineering

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Time-Series Forecasting, Bayesian Optimization (TuRBO, SAASBO), CMA-ES, Computer Vision (YOLO, OpenCV), Collaborative Filtering, A/B Testing

Data Engineering & Analytics: Microsoft Fabric, Medallion Architecture, Delta Lake, PySpark, Spark SQL, SCD Type 2, Data Quality & Reconciliation, Airflow, DVC, PostgreSQL, SQLite, DynamoDB

Backend & Orchestration: FastAPI, Fastify, Temporal (durable workflows), REST/SSE APIs, Next.js 15, React, Drizzle ORM, Stripe, Clerk

Testing & QA: Vitest, Cypress, Cucumber (BDD), Page Object Model, Allure, Unit / Integration / E2E, CI Gating

Cloud & DevOps: AWS (SageMaker, Lambda, API Gateway, S3, DynamoDB, CloudFront, SAM), Azure (Fabric, Entra ID), GCP, Docker, GitHub Actions, MLflow, Prometheus, Grafana, Datadog RUM, Vercel, Linux, Git

Work Experience

Handshake / *AI Evaluation Specialist* **July 2025**

- **Situation:** Frontier AI labs need agentic benchmark tasks that measure real terminal skill and cannot be gamed by the model under test.
- **Action:** Contributed 6+ Docker-sandboxed agentic-AI evaluation tasks to Terminal-Bench 2 (Harbor), enforcing four-way consistency and a tamper-proof verifier design.
- **Result:** Achieved an 88.9% acceptance rate on frontier-model benchmarking tasks, meeting the GPT-5.4 acceptance bar.
- **Tech Stack:** Docker, Python, Agentic AI, Evaluation Frameworks, LLM Benchmarking

Self Employed / *Freelance Full Stack Developer Repo Link* **Feb 2025 – May 2025**

- **Situation:** Morsaab's, a vegetarian restaurant in New Delhi, needed online ordering, bookings and an editable menu.
- **Action:** Built a Next.js 15 + TypeScript platform: a 9-category menu on Neon Postgres + Drizzle, Stripe Checkout with signed webhooks, a Clerk-gated admin CRM, Vercel Blob and Datadog RUM; mapped an AWS SAM serverless path (Lambda, DynamoDB, CloudFront).
- **Result:** Shipped 9.8k lines in under 40 hours, about 4x faster than a typical 3–4 week build; live on Vercel.

WBSEDCL / *Architect-Machine Learning Trainee Repo* **Mar 2024 – Apr 2024**

- **Situation:** The state utility read electricity meters by hand, which was slow and error-prone at scale.
- **Action:** Built an automatic meter-reading pipeline that crops the meter display, then benchmarked YOLO v5, v7 and v9-e for digit recognition; cleaned 466 images with erosion and HSV correction.
- **Result:** Reached 99% accuracy, 94% precision and 93% mAP, cut reading errors by 95% and saves a projected 500+ staff-hours a year.
- **Tech Stack:** Python, YOLO v5 / v7 / v9-e, Roboflow, OpenCV

NIELIT Kolkata / *Artificial Intelligence Intern Repo* **Sep 2023 – Mar 2024**

- **Situation:** Manual fruit grading is slow and inconsistent, as fruit of one variety looks nearly identical.
- **Action:** Designed a real-time, non-destructive grading framework comparing AlexNet and ResNet v2 across two datasets, with segmentation-focused preprocessing feeding a conveyor-belt sorting prototype.
- **Result:** 99% accuracy on 2.2K images and 98.33% (0.9752 F1) on 15.5K images; published at IEEE.
- **Tech Stack:** Python, Keras, NumPy, OpenCV, AlexNet, ResNet v2

Projects

NovaCart — Microsoft Fabric Medallion Pipeline *(team of 5)* *Repo* **Data Engineering**

- **Situation:** Finance at a 5-country retailer got conflicting revenue figures; late corrections could overwrite newer orders.
- **Action:** Built a bronze–silver–gold Fabric lakehouse: newer-wins guarded Delta MERGEs, SCD Type 2 customers with point-in-time joins, as-of FX conversion, typed quarantine and a retrying Data Pipeline.
- **Result:** Merged 2,963 order versions into 945 orders, blocked 10 stale updates and matched $295,585.57 of revenue to the cent with an independent check; re-runs change 0 rows and all 224 tests pass, on about $2–3 of compute.
- **Tech Stack:** Microsoft Fabric, PySpark, Spark SQL, Delta Lake, Data Pipelines, Entra ID

SmartDialer — Outbound-Dialer Simulator with Safety Engine *Repo* *Link* **QA / Systems**

- **Situation:** A predictive dialer that over-dials must never dial a do-not-call number or leak capacity in an outage.
- **Action:** Built a TypeScript engine where every dial passes 13 ordered, explainable safety rules, with a variance-guarded pacer, timeout watchdogs and a SHA-256 determinism digest over seeded runs.
- **Result:** 529 Vitest tests and 35 Cypress–Cucumber BDD scenarios gate every push in CI with 0 do-not-call breaches; a covering index made contact reservation 40x faster at 1,500 agents (722 to 17.9 μs).
- **Tech Stack:** TypeScript, Fastify, React, SQLite, Vitest, Cypress, Cucumber, Allure, GitHub Actions

Order Supervisor — Long-Running AI Agent on Temporal *Repo* **Agentic AI / FDE**

- **Situation:** Orders need attention for days (failed payments, delays), but an always-on LLM loop is costly and fragile.
- **Action:** Gave each order a durable Temporal workflow whose agent wakes on 3 triggers; a rule-based classifier gates events by urgency, and the agent acts through 6 tools with rolling memory.
- **Result:** A 2.3k-line full-stack POC with 10 REST endpoints that sleeps on durable timers instead of polling, caps history via continue-as-new after 200 turns, and runs keyless via a deterministic mock policy.
- **Tech Stack:** Python, Temporal, FastAPI, PostgreSQL, SQLAlchemy, Anthropic Claude API, Next.js, Tailwind CSS

EmpaChat — Empathetic Response AI *Repo* **LLM Fine-Tuning**

- **Situation:** Chatbots answer distressed users in a flat tone, and fully fine-tuning a 1.5B model exceeded one GPU.
- **Action:** Fine-tuned GPT-2 XL with LoRA (r = 8) and 4-bit QLoRA on 58K GoEmotions comments as an emotion classifier plus an emotion-conditioned generator, served via FastAPI and Gradio.
- **Result:** Trained 8.4M of 1.56B parameters (a 99.4% reduction) for about 89% classification accuracy at about 150 ms GPU latency, and a 35% preference gain over the baseline in a simulated A/B test.
- **Tech Stack:** PyTorch, HuggingFace Transformers, PEFT (LoRA/QLoRA), FastAPI, Gradio, Docker

Basket Trading Optimization Framework *Repo* **Quant ML**

- **Situation:** Johansen-test cointegration weights look good in-sample but often lose money on unseen data.
- **Action:** Extended a Bayesian-optimization baseline into a 6-optimizer benchmark (CVFS-CMA-ES, BO, TuRBO, SAASBO, CMA-ES) maximising out-of-sample Sharpe after 0.1% costs.
- **Result:** CVFS-CMA-ES reached a 3.851 Sharpe ratio (9x standard CMA-ES's 0.420) with only –3.99% max drawdown, while Bayesian optimization produced the highest total return at 63.76%.
- **Tech Stack:** Python, scikit-optimize, pycma (CMA-ES), Statsmodels, SciPy, Pandas, yfinance

Bookie — End-to-End MLOps Book Recommender *Repo* **MLOps**

- Packaged a full recommender lifecycle into one `docker compose up`: a 4-stage Airflow DAG (ingest, preprocess, train SVD collaborative filtering, evaluate) with DVC data versioning, MLflow experiment tracking, FastAPI serving, a React front end and Prometheus/Grafana monitoring — 6 services in total.
- **Tech Stack:** Python, Airflow, DVC, MLflow, FastAPI, React, Docker, Prometheus, Grafana

Research Work

Deep Neural Network Based Yield Phenotyping Framework: Fruit Grading and Classification, *IEEE*
Authors: Cassandra Chang, Sayantan Sikdar | *Link*

Certifications and Achievements

Google Cloud Training: Cloud Engineering Fundamentals & Essentials, Cloud Application Development, Cloud ML-AI, DevOps & Kubernetes | Anthropic: Claude 101, Claude Code 101 | NVIDIA: Fundamentals of Deep Learning | AWS Machine Learning Scholarship 2021, Udacity – Recipient | GATE CS 2024 – Qualified | Youngest IEEE Student Researcher